

CSC 211: Computer Programming

Arrays, Vectors

Dr. Michael Conti

Department of Computer Science and Statistics
University of Rhode Island

Summer 2026



Original design and development by Dr. Marco Alvarez

Question

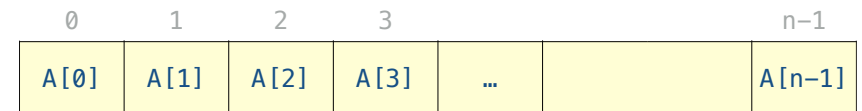
- Write a program that reads in **3** values and outputs the same values in reverse order
- Write a program that reads in **n** values and outputs the same values in reverse order

2

Arrays

Arrays

- An array is a **contiguous** sequence of elements of the **same type**
- Each element (data in array) can be accessed using its **index**
array name: **A**
array length: **n**



all elements of the same data type

4

Declaration

```
// array declaration by specifying size
int myarray1[100];

// can also declare an array of
// user specified size
int n = 8;
int myarray2[n];

// can declare and initialize elements
double arr[] = { 10.0, 20.0, 30.0, 40.0 };
// compiler figures the right size

// a different way
int arr[5] = { 1, 2, 3 };
// compiler creates an array of length 5 and
// initializes first 3 elements
```

5

Initialization and indexing

- Elements in an array **must be initialized** before use
 - otherwise, their initial values are **undetermined**
 - can use a loop to initialize values
- Individual elements can be accessed by using the **subscription operator []**

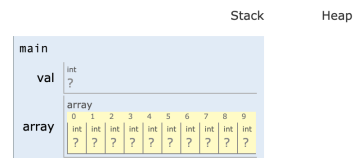
```
int array[4];
array[0] = 5;
array[1] = array[0] + 10;
array[2] = array[1] + 20;
array[3] = array[2] + 30;
```

0	1	2	3
5	15	35	65

6

Trace the code

```
C++ (gcc 4.8, C++11)
EXPERIMENTAL! known bugs/limitations
1 int main() {
2   int val = 0;
3   int array[10];
4
5   for (int i = 0 ; i < 10 ; i++) {
6     val += 50;
7     array[i] = val;
8   }
9
10  return 0;
11 }
```



7

Out of bounds?

- There is no **out of bounds** checking at compile time
 - unexpected output**

A[9] ?

0	1	2	3	4	5	6	7		
?	?	?	10	20	50	100	70	50	30



8

What is the output?

```
#include <iostream>

int main() {
    int myarray[5];
    for (int i = 0 ; i < 5 ; i++) {
        myarray[i] = i;
    }
    for (int i = -10 ; i < 10 ; i++) {
        std::cout << myarray[i] << ' ';
    }
    std::cout << '\n';
    return 0;
}
```

9

Computer memory



- ✓ A **memory address** is a reference to a specific memory location
- ✓ Memory addresses are **fixed-length** sequences of digits (hexadecimal codes)
- ✓ **Word**-oriented memory organization (word size 32-bit in this illustration)

0x00000000	
0x00000004	
0x00000008	
0x0000000C	
0x00000010	
0x00000014	
0x00000018	
...	
...	
...	
0xFFFFFEC	
0xFFFFF0	
0xFFFFF4	
0xFFFFF8	
0xFFFFFC	

address content

https://en.wikipedia.org/wiki/Random-access_memory

10

Computer memory (example)

```
int main() {
    int a = 4;
    int i = 0;
    double b = 10;
    int arr[5];

    for (; i < 5 ; i++) {
        arr[i] = i * 100;
    }

    return 0;
}
```

Assuming 32-bit words

...	
0x91340A04	
0x91340A08	4
0x91340A0C	5
0x91340A10	
0x91340A14	10
0x91340A18	0
0x91340A1C	100
0x91340A20	200
0x91340A24	300
0x91340A28	400
0x91340A2C	
0x91340A30	
0x91340A34	
...	

11

Passing arrays to functions

- When specifying the parameter, use **empty brackets**
- When providing the argument, use the **array name**
 - ✓ need to pass the **array length** separately

```
void zeros(int a[], int n) {
    for (int i = 0 ; i < n ; i++) {
        a[i] = 0;
    }
}

int main() {
    int array[5];
    zeros(array, 5);
    // do stuff
}
```

12

Base address

- **Base address** is the memory location of the first element in an array
 - ✓ base address of **arr** is **0x91340A18** (previous example)

- When passing arrays to functions, the base address of the array is passed to the formal parameter

...	
0x91340A04	
0x91340A08	4
0x91340A0C	5
0x91340A10	10
0x91340A14	
0x91340A18	0
0x91340A1C	100
0x91340A20	200
0x91340A24	300
0x91340A28	400
0x91340A2C	
0x91340A30	
0x91340A34	
...	

13

Base address

Users > michaelconti > Desktop > G+ arr.cpp

```
2
3 int main(){
4
5     int array[4];
6     array[0] = 5;
7     array[1] = array[0] + 10;
8     array[2] = array[1] + 20;
9     array[3] = array[2] + 30;
10
11     std::cout << array;
12
13     return 0;
14 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

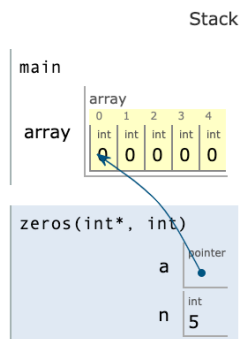
```
michaelconti@Michaels-MacBook-Pro-2 Desktop % ./temp
0x16ef638a8%
michaelconti@Michaels-MacBook-Pro-2 Desktop %
```

14

Passing arrays to functions

C++ (gcc 4.8, C++11)
EXPERIMENTAL! [known bugs/limitations](http://pythontutor.com/cpp.html)

```
1 void zeros(int a[], int n) {
2     for (int i = 0 ; i < n ; i ++ ) {
3         a[i] = 0;
4     }
5 }
6
7 int main() {
8     int array[5];
9     zeros(array, 5);
10    // do stuff
11 }
```



15

Vectors

Vectors

- Data structure for organizing elements

- **#include <vector>**

```
// declare
std::vector<int> myVector;

// initializer list (c++17)
std::vector<int> vector1 = {1, 2, 3, 4, 5};
```

17

Declaration

```
// declare
std::vector<int> myVector;

// initializer list (c++17)
std::vector<int> vector1 = {1, 2, 3, 4, 5};
```

18

Important Methods

```
//declare vector without size
std::vector<int> myVector;

//declare vector with size
std::vector<int> myVector(20);

//add element into vector
myVector.push_back(5);

//add element into vector
myVector[0] = 5;

//access vector (with bound checking)
myVector.at(0);

//access vector (without bound checking)
myVector[0];

//change vector element
myVector[0] = 10;

//remove element into vector
myVector.pop_back();
```

19

Question

- Write a function that receives an array of integers and reverses the contents of the array

20

Question

- Write a function that receives an array and returns the smallest element in that array.